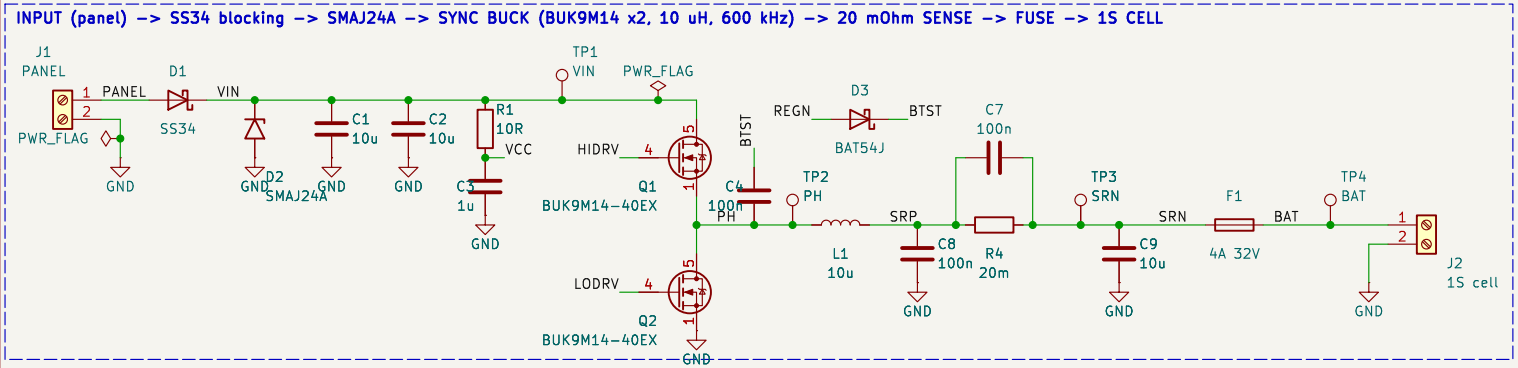
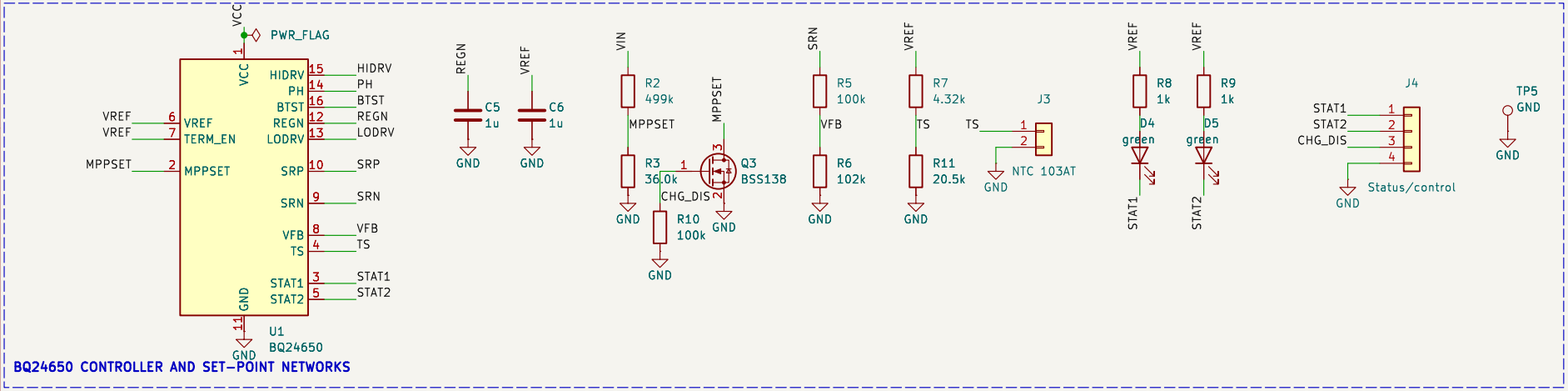
WARNING: Reference design. Test first with a current-limited supply and protection. Not for unattended use.

Design only, not fabricated, not bench-tested. Li-ion: use a cell or pack with its own protection circuit and a 103AT NTC on J3.





KEY VALUES (README has the derivations and sources)

Charge voltage = 2.1 V x (1 + 100k/102k) = 4.159 V; worst case 4.134-4.194 V incl. VFB leakage (never above 4.20 V).

Charge current = 40 mV / 20 mOhm = 2.0 A (+-3 %); precharge and termination 0.2 A (+-25 %). Cycle-by-cycle limit 4 A.

MPP: VCC-side regulation = 1.2 V x (1 + 499k/36.0k) = 17.83 V (panel ~18.2 V with D1). Set to your panel's Vmp at operating temperature.

L1 10 uH: ripple <= 0.83 A worst case (41 % of 2 A). Output LC f0 = 15.9 kHz with 10 uF (datasheet 12-17 kHz).

TS: R7 4.32k / R11 20.5k with 103AT (B-model): charge 0-45 C, cut-off ~49 C. Recompute from the Semitec R-T table.

SMAJ24A clamps at 38.9 V at full pulse, above BQ24650 VCC abs max 33 V. Panel Voc must stay <= 22 V.

CHG_DIS high pulls MPPSET below 75 mV and stops charging. STAT1 low = charging, STAT2 low = done.

Generated from stock KiCad 9 library symbols by scripts/gen_schematic.py

Hardware licence: CERN-OHL-P-2.0

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Abdulazeez Bello

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File: solar-charger.kicad_sch

Title: Solar charger: BQ24650 for one Li-ion cell

Size: A4	Date: 2026-10-18	**Rev: A**
KiCad E.D.A. 9.0.3		Id: 1/1